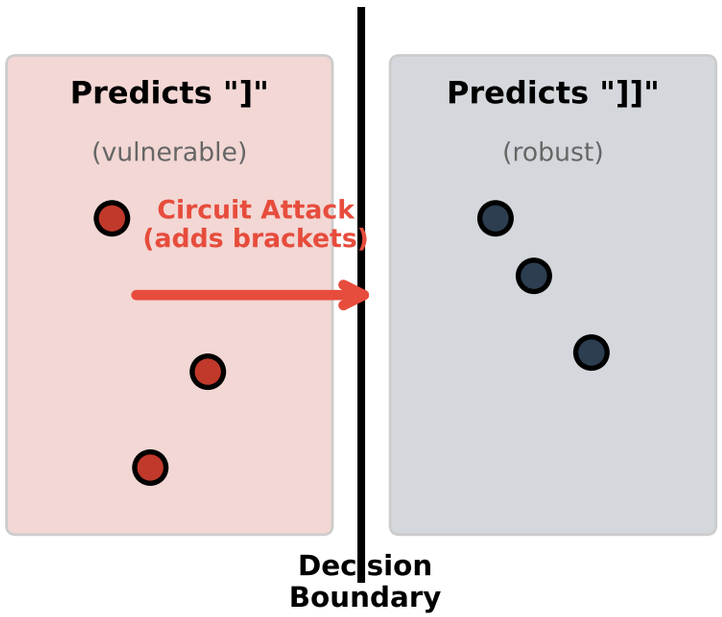
Directional Collision Principle

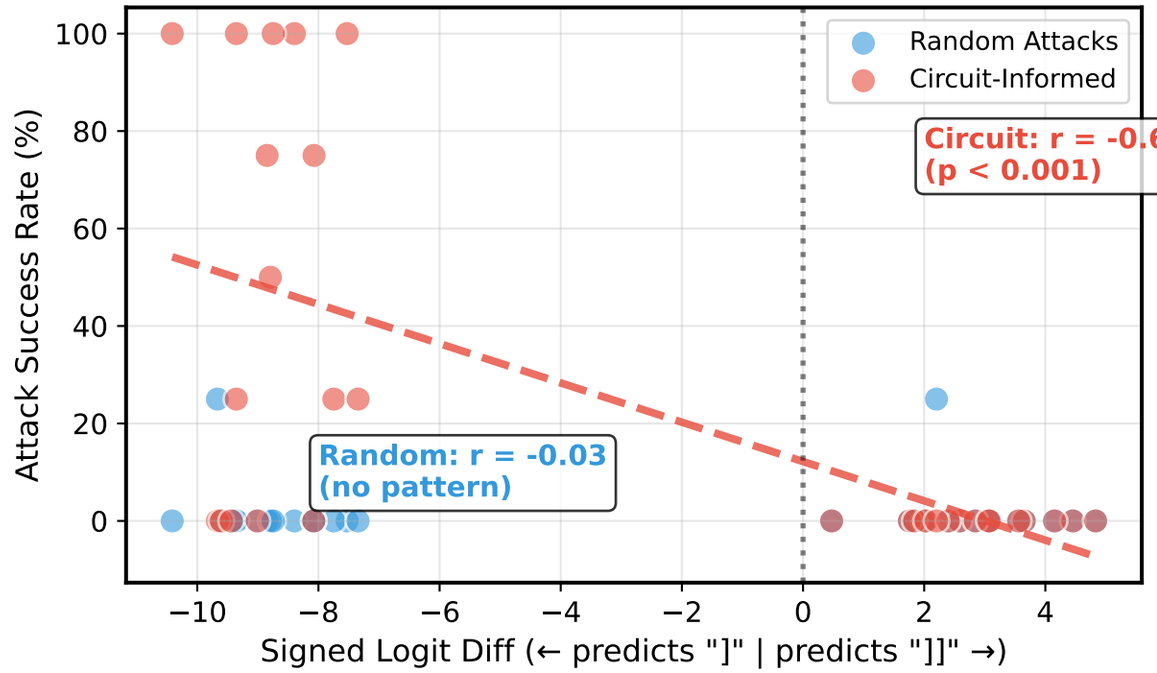
A Geometric Theory of Adversarial Vulnerability in Neural Networks

KEY FINDING: Theory-based prediction achieves 84% accuracy with 100% recall in identifying vulnerable inputs

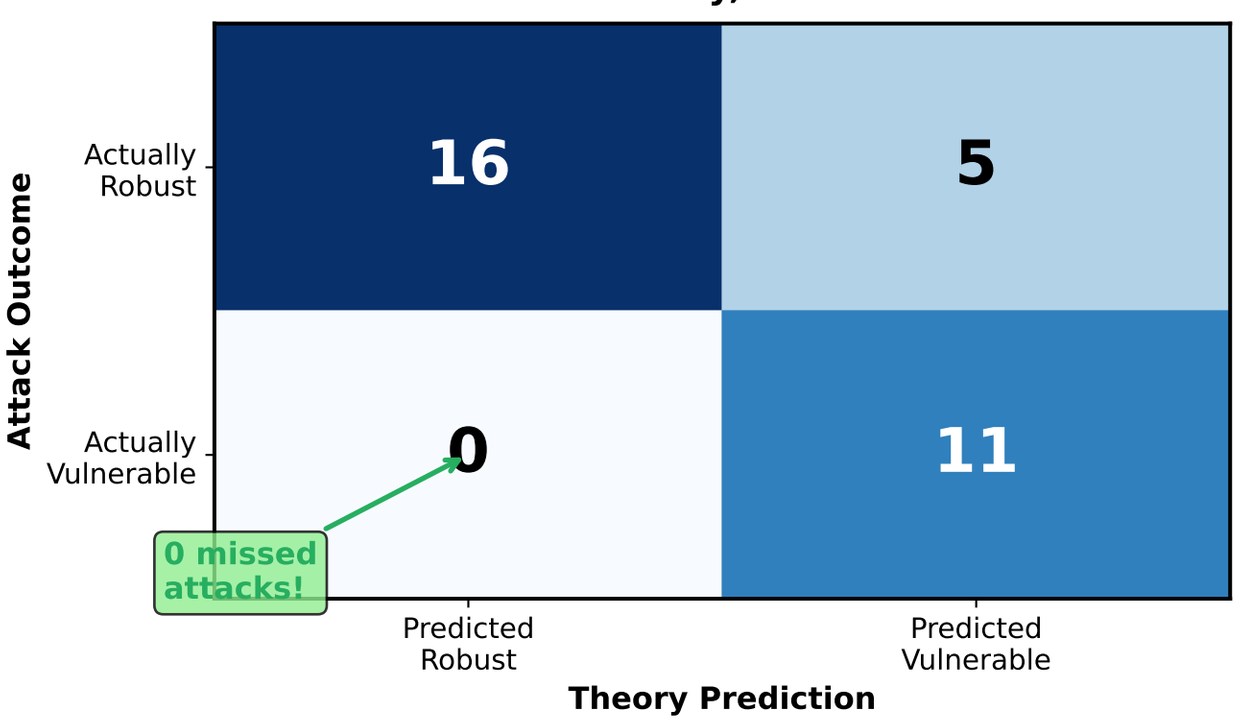
A. Directional Collision Principle
Attacks push toward "]]"; only "]" samples can flip



B. Directional Validation
Strong negative correlation for targeted attacks only



C. Prediction Accuracy
84% accuracy, 100% recall



D. Key Results

16×
Attack Efficiency Improvement
Circuit-informed vs Random

84%
Prediction Accuracy
27/32 samples correct

100%
Attack Recall
Caught all 11 attacks

r = -0.62
Directional Correlation
$p < 0.001$